A. General Information

1	Project Name	Cesar Chavez Apartments			
2	Run Title	Compliance			
3	Project Location	3720 E Cesar Chavez Blvd			
4	City	Fresno	**5**	Standards Version	Compliance 2022
6	Zip code	93702	**7**	Compliance Software (version)	CBECC 2022.3.2-SP1 (1369)
8	Climate Zone	13	**9**	Building Orientation (deg)	0
10	Building Type(s)	• High-Rise Residential	**11**	Weather File	FRESNO-YOSEMITE_STYP20.epw
12	Project Scope	• New complete scope	**13**	Number of Dwelling Units	54
14	Total Conditioned Floor Area in Scope (ft^2)	50722	**15**	Total # of hotel/motel rooms	0
16	Total Unconditioned Floor Area (ft^2)	16595	**17**	Fuel Type	None
18	Nonresidential Conditioned Floor Area	0	**19**	Total # of Stories (Habitable Above Grade)	4
20	Residential Conditioned Floor Area	50722			

B. PROJECT SUMMARY

Table B shows which building components are included in the performance calculation. If indicated as not included, the project must show compliance prescriptively if within the permit application.

Building Components Complying via Performance						**Building Components Complying Prescriptively**	
Envelope (See Table G)	Nonres	Not Included	Solar Thermal Water Heating (See Table I3)	☐	Performance	The following building components are ONLY eligible for prescriptive compliance and should be documented on the NRCC form listed if within the scope of the permit application (i.e. compliance will not be shown on the NRCC-PRF-E).	
	MultiFam	Performance		☒	Not Included		
Mechanical (See Table H)	Nonres	Not Included	Covered Process: Commercial Kitchens (see Table J)	☐	Performance	Indoor Lighting (Unconditioned) 140.6 & 170.2(e)	NRCC-LTI-E is required
	MultiFam	Performance		☒	Not Included	Outdoor Lighting 140.7 & 170.2(e)	NRCC-LTO-E is required
Domestic Hot Water (See Table I)	Nonres	Not Included	Covered Process: Laboratory Exhaust (see Table J)	☐	Performance	Sign Lighting 140.8 & 170.2(e)	NRCC-LTS-E is required
	MultiFam	Performance		☒	Not Included	**Building Components Complying with Mandatory Measures**	
Lighting (Indoor Conditioned, see Table K)	Nonres	Not Included	Photovoltaics (see Table F)	☒	Performance	Electrical power systems, commissioning, solar ready, elevator and escalator requirements are mandatory and should be documented on the NRCC form listed if applicable (i.e. compliance will not be shown on the NRCC-PRF-E.)	
	MultiFam	Performance		☐	Not Included	Electrical Power Distribution 110.11	NRCC-ELC-E is required
			Battery (see Table F)	☐	Performance	Commissioning 120.8	NRCC-CXR-E is required
				☒	Not Included	Solar and Battery 110.10	NRCC-SAB-E is required

C1. COMPLIANCE SUMMARY

COMPLIES[3]			
	Time Dependent Valuaton (TDV)		Source Energy Use
	Efficiency[1] (kBtu/ft^2 - yr)	Total[2] (kBtu/ft^2 - yr)	Total[2] (kBtu/ft^2 - yr)
Standard Design	146.26	75.24	8.43
Proposed Design	124.25	-81.12	3.8
Compliance Margins	**22.01**	**156.36**	**4.63**
	Pass	**Pass**	**Pass**

[1] Efficiency measures include improvements like a better building envelope and more efficient equipment

[2] Compliance Totals include efficiency, photovoltaics and batteries

[3] New Construction, Complete Addition Scope: Building complies when all efficiency and total compliance margins are greater than or equal to zero and unmet load hour limits are not exceeded
Existing, Addition and Alteration Scope: Building complies when efficiency compliance margin is greater than or equal to zero and unmet load hour limits are not exceeded

C2. TDV ENERGY COMPLIANCE RESULTS FOR PERFORMANCE COMPONENTS (Annual TDV Energy Use, kBtu/ft^2 - yr)			
COMPLIES[2]			
Energy Component	**Standard Design (TDV)**	**Proposed Design (TDV)**	**Compliance Margin (TDV)[1]**
Space Heating	5.66	5.72	-0.06
Space Cooling	49.71	47.91	1.8
Indoor Fans	22.07	19.51	2.56
Heat Rejection	0	0	0
Pumps & Misc.	0.79	0.75	0.04
Domestic Hot Water	37.72	20.05	17.67
Indoor Lighting	30.31	30.31	0
Flexibility	---	---	---
EFFICIENCY COMPLIANCE TOTAL	**146.26**	**124.25**	**22.01 (15%)**
Photovoltaics	-69.31	-205.37	136.06
Batteries	-1.71	---	-1.71
TOTAL COMPLIANCE	**75.24**	**-81.12**	**156.36 (207.8%)**

[1] *Notes: This number in parenthesis following the Compliance Margin in column 4, represents the Percent Better than Standard.*

C3. TDV ENERGY RESULTS FOR NON-REGULATED COMPONENTS[1]			
Non-Regulated Energy Component	**Standard Design (TDV)**	**Proposed Design (TDV)**	**Compliance Margin (TDV)[1]**
Receptacle	58.55	58.55	---
Process	59.52	59.5	0.02
Other Ltg	8.43	8.43	---
Process Motors	---	---	---
TOTAL (TOTAL COMPLIANCE + NON-REGULATED COMPONENTS)	**201.74**	**45.36**	**156.38 (77.5%)**
[1] *Notes: This table is not used for Energy Code Compliance.*			

C4. SOURCE ENERGY COMPLIANCE RESULTS FOR PERFORMANCE COMPONENTS (Annual SOURCE Energy Use, kBtu/ft^2 /yr)

COMPLIES[2]

Energy Component	Standard Design (SOURCE)	Proposed Design (SOURCE)	Compliance Margin (SOURCE)[1]
Space Heating	0.77	0.78	-0.01
Space Cooling	2.24	2.19	0.05
Indoor Fans	1.69	1.47	0.22
Heat Rejection	0	0	0
Pumps & Misc.	0.1	0.1	0
Domestic Hot Water	3.51	1.78	1.73
Indoor Lighting	2.75	2.75	0
Flexibility	---	---	---
EFFICIENCY COMPLIANCE TOTAL	**11.06**	**9.07**	**1.99 (18%)**
Photovoltaics	-1.89	-5.27	3.38
Batteries	-0.74	---	-0.74
TOTAL COMPLIANCE	**8.43**	**3.8**	**4.63 (54.9%)**

[1] *Notes: This number in parenthesis following the Compliance Margin in column 4, represents the Percent Better than Standard.*

C5. SOURCE ENERGY RESULTS FOR NON-REGULATED COMPONENTS[1]			
Non-Regulated Energy Component	Standard Design (SOURCE)	Proposed Design (SOURCE)	Compliance Margin (SOURCE)[1]
Receptacle	5.4	5.4	---
Process	4.61	4.61	---
Other Ltg	0.85	0.85	---
Process Motors	---	---	---
TOTAL (TOTAL COMPLIANCE + NON-REGULATED COMPONENTS)	**19.29**	**14.66**	**4.63 (24%)**
[1] *Notes: This table is not used for Energy Code Compliance.*			

C6. 'ABOVE CODE' QUALIFICATIONS	
☐ This project is pursuing CalGreen Tier 1	☐ This project is pursuing CalGreen Tier 2

C7. ENERGY USE SUMMARY

Energy Component	Standard Design Site (MWh)	Proposed Design Site (MWh)	Margin (MWh)	Standard Design Site (MBtu)	Proposed Design Site (MBtu)	Margin (MBtu)
Space Heating	9.6	9.7	-0.1	---	---	---
Space Cooling	60.9	58.5	2.4	---	---	---
Indoor Fans	36.5	31.7	4.8	---	---	---
Heat Rejection	---	---	---	---	---	---
Pumps & Misc.	1.4	1.3	0.1	---	---	---
Domestic Hot Water	72	42.7	29.3	---	---	---
Indoor Lighting	57.5	57.5	0	---	---	---
Flexibility	---	---	---	---	---	---
EFFICIENCY TOTAL	**237.9**	**201.4**	**36.5**	**0**	**0**	**0**
Photovoltaics	-177.7	-513.7	336	---	---	---
Batteries	1.4	---	---	---	---	---
ENERGY USE SUBTOTAL	**61.6**	**-312.3**	**373.9**	**0**	**0**	**0**
Receptacle	108.3	108.3	0	---	---	---
Process	114.7	114.6	0.1	---	---	---
Other Ltg	13.9	13.9	0	---	---	---
Process Motors	---	---	---	---	---	---
ENERGY USE TOTAL	**298.5**	**-75.5**	**374**	**0**	**0**	**0**

C8. ENERGY USE INTENSITY (EUI)				
	Standard Design (kBtu/ft^2 / yr)	**Proposed Design (kBtu/ft^2 / yr)**	**Margin (kBtu/ft^2 / yr)**	**Margin Percentage**
GROSS EUI[1]	24.14	22.21	1.93	8
NET EUI[1]	15.13	-3.83	18.96	125.31

[1] *Notes: Gross EUI is Energy Use Total (not including PV)/Total Building Area. Net EUI is Energy Use Total (including PV)/Total Building Area.*

D1. EXCEPTIONAL CONDITIONS

- PV/Battery Building Type has been modified from software defaults for one or more spaces. Review project’s PV/Battery Building Type(s) with documentation author. Refer to Energy Code section 140.10 for Nonresidential or 170.2(g) for more information.

D2. MULTIFAMILY REQUIRED SPECIAL FEATURES

- IAQ Ventilation System: supply outside air inlet, filter, and H/ERV cores accessible per RACM Reference Manual
- IAQ Ventilation System: fault indicator display
- Central Heat Pump Water Heater
- Multifamily: Recirculating with no control (continuous pumping)

F1. REQUIRED PV SYSTEMS

01	02	03	04	05	06	07	08	09	10	11	12
DC System Size (kWdc)	**Exception[1]**	**Module Type**	**Array Type**	**Power Electronics**	**CFI**	**Azimuth (deg)**	**Tilt Input**	**Array Angle (deg)**	**Tilt: (x in 12)**	**Inverter Eff. (%)**	**Annual Solar Access (%)**
350		Premium (~18-20%)	Fixed	none	false	180	Degrees	5	1.05	96	98

[1]*See Table D1 for any PV exceptions used.*

F1B. PV BATTERY BUILDING TYPE(S)		
01	02	03
Building Occupancy Type* (From Table 140.10-A/B and 170.2-U/V)	Conditioned Floor Area (ft^2)	Unconditioned Floor Area (ft^2)
Grocery	0	0
High-Rise Multifamily	50722	16595
Office, Financial Institutions, Unleased Tenant Space	0	0
Retail	0	0
School	0	0
Warehouse	0	0
Auditorium, Convention Center, Hotel/Motel, Library, Medical Office Building/Clinic, Restaurant, Theater	0	0
None	0	0

**Building Occupancy Types are defined in Section 100.1 of the Energy Code*

F3. DWELLING UNIT INFORMATION			
01	02	03	04
Dwelling Unit Name	Dwelling Unit Type	Zone	Zone Group Multiplier
A1_L01_1BR-(1/4)	A1	Dwelling Unit_L01_1BR	1
A1_L01_1BR-(2/4)	A1	Dwelling Unit_L01_1BR	1
A1_L01_1BR-(3/4)	A1	Dwelling Unit_L01_1BR	1
A1_L01_1BR-(4/4)	A1	Dwelling Unit_L01_1BR	1
B1_L01_2BR-(1/4)	B1	Dwelling Unit_L01_2BR	1
B1_L01_2BR-(2/4)	B1	Dwelling Unit_L01_2BR	1
B1_L01_2BR-(3/4)	B1	Dwelling Unit_L01_2BR	1
B1_L01_2BR-(4/4)	B1	Dwelling Unit_L01_2BR	1
C1_L01_3BR-(1/4)	C1	Dwelling Unit_L01_3BR	1
C1_L01_3BR-(2/4)	C1	Dwelling Unit_L01_3BR	1
C1_L01_3BR-(3/4)	C1	Dwelling Unit_L01_3BR	1
C1_L01_3BR-(4/4)	C1	Dwelling Unit_L01_3BR	1
A1_L02_1BR-(1/4)	A1	Dwelling Unit_L02-03_1BR	2
A1_L02_1BR-(2/4)	A1	Dwelling Unit_L02-03_1BR	2
A1_L02_1BR-(3/4)	A1	Dwelling Unit_L02-03_1BR	2
A1_L02_1BR-(4/4)	A1	Dwelling Unit_L02-03_1BR	2
B1_L02_2BR-(1/5)	B1	Dwelling Unit_L02-03_2BR	2
B1_L02_2BR-(2/5)	B1	Dwelling Unit_L02-03_2BR	2
B1_L02_2BR-(3/5)	B1	Dwelling Unit_L02-03_2BR	2
B1_L02_2BR-(4/5)	B1	Dwelling Unit_L02-03_2BR	2
B1_L02_2BR-(5/5)	B1	Dwelling Unit_L02-03_2BR	2
C1_L02_3BR-(1/5)	C1	Dwelling Unit_L02-03_3BR	2
C1_L02_3BR-(2/5)	C1	Dwelling Unit_L02-03_3BR	2
C1_L02_3BR-(3/5)	C1	Dwelling Unit_L02-03_3BR	2
C1_L02_3BR-(4/5)	C1	Dwelling Unit_L02-03_3BR	2
C1_L02_3BR-(5/5)	C1	Dwelling Unit_L02-03_3BR	2

F3. DWELLING UNIT INFORMATION

01	02	03	04
Dwelling Unit Name	Dwelling Unit Type	Zone	Zone Group Multiplier
A1_L02-2_1BR-(1/4)	A1	Dwelling Unit_L04_1BR	1
A1_L02-2_1BR-(2/4)	A1	Dwelling Unit_L04_1BR	1
A1_L02-2_1BR-(3/4)	A1	Dwelling Unit_L04_1BR	1
A1_L02-2_1BR-(4/4)	A1	Dwelling Unit_L04_1BR	1
B1_L02-2_2BR-(1/5)	B1	Dwelling Unit_L04_2BR	1
B1_L02-2_2BR-(2/5)	B1	Dwelling Unit_L04_2BR	1
B1_L02-2_2BR-(3/5)	B1	Dwelling Unit_L04_2BR	1
B1_L02-2_2BR-(4/5)	B1	Dwelling Unit_L04_2BR	1
B1_L02-2_2BR-(5/5)	B1	Dwelling Unit_L04_2BR	1
C1_L02-2_3BR-(1/5)	C1	Dwelling Unit_L04_3BR	1
C1_L02-2_3BR-(2/5)	C1	Dwelling Unit_L04_3BR	1
C1_L02-2_3BR-(3/5)	C1	Dwelling Unit_L04_3BR	1
C1_L02-2_3BR-(4/5)	C1	Dwelling Unit_L04_3BR	1
C1_L02-2_3BR-(5/5)	C1	Dwelling Unit_L04_3BR	1

F4. DWELLING UNIT TYPES

01	02	03	04	05	06	07
Name	**CFA (ft^2)**	**Number of Bedrooms**	**Number in Building**	**Space Conditioning Systems Assigned**	**DHW System Name**	**IAQ Vent Fan Name**
A1	554	1	16	A1 | :MAESTRO OLIMPIA PRO12 HP #01925:::2:3	ResidentialDHWSystem 1	Central Supply
B1	807	2	19	B1 | :MAESTRO OLIMPIA PRO12 HP #01925:::2:3	ResidentialDHWSystem 1	Central Supply
C1	1254	3	19	C1 | :MAESTRO OLIMPIA PRO12 HP #01925:::2:3	ResidentialDHWSystem 1	Central Supply

G1. ENVELOPE GENERAL INFORMATION (conditioned spaces only)

01	02	03	04
Opaque Surfaces & Orientation	**Total Gross Surface Area (ft^2)**	**Total Fenestration Area (ft^2)**	**Window to Wall Ratio (%)**
North-Facing[1]	630	30	4.76
East-Facing[2]	11391.8	3720	32.66
South-Facing[3]	4189.05	451.19	10.77
West-Facing[4]	18243.5	3960	21.71
Total	**34454.4**	**8161.19**	**23.69**
Roof	15218	0	0

Notes

[1]North-Facing is oriented to within 45 degrees of true north, including 4500'00" east of north (NE), but excluding 4500'00" west of north (NW),

[2]East-Facing is oriented to within 45 degrees of true east, including 4500'00" south of east (SE), but excluding 4500'00" north of east (NE),

[3]South-Facing is oriented to within 45 degrees of true south, including 4500'00" west of south (SW), but excluding 4500'00" east of south (SE),

[4]West-Facing is oriented to within 45 degrees of true west, including 4500'00" north of west (NW), but excluding 4500'00" south of west (SW),

G2B. ROOFING PRODUCT SUMMARY (MULTIFAMILY AND COMMON AREAS)					
01	02	03	04	05	06
Name	Roof Pitch	Roof Rise (x in 12)	Aged Solar Reflectance	Thermal Emittance	SRI
Roof (Front Combined) : GYM_L01	Low slope	0	0.1	0.85	N/A
ResidentialCathedralCeiling 1-2-2-2	Low slope	0	0.1	0.85	N/A
ResidentialCathedralCeiling 1-2-2	Low slope	0	0.1	0.85	N/A
ResidentialCathedralCeiling 1_1BR	Low slope	0	0.1	0.85	N/A
ResidentialCathedralCeiling 1_2BR	Low slope	0	0.1	0.85	N/A
ResidentialCathedralCeiling 1_3BR	Low slope	0	0.1	0.85	N/A
ResidentialCathedralCeiling 1-2	Low slope	0	0.1	0.85	N/A

G5. OPAQUE SURFACE ASSEMBLY SUMMARY

01	02	03	04	05	06		07	08	09	10
Surface Name	Construction Type	Area (ft^2)	Framing Type	Cavity R-Value	Continuous R-Value		Units	Value	Description of Assembly Layers	Status[1]
					Interior	Exterior				
R-21 Exterior Wall	Exterior Walls	35,561.4	Wood Framed Wall	21	0	3	U-factor	0.0543	Inside Finish: Gypsum Board Cavity / Frame: R-21 / 2x6 Sheathing / Insulation: R-3 Sheathing Exterior Finish: Synthetic Stucco	N
Rigid + Cavity Roof insulation	Cathedral Ceilings	17,878	Wood Framed Ceiling	19	0	30	U-factor	0.0188	Roofing: Light Roof (Asphalt Shingle) Above Deck Insulation: R-30 Sheathing Roof Deck: Wood Siding/sheathing/decking Radiant Barrier Cavity / Frame: R-19 / 2x12 Sheathing / Insulation: Wood Siding/sheathing/decking Inside Finish: Gypsum Board	N
R-13 Interior Wall	Interior Walls	23,639.8	Wood Framed Wall	13	0	0	U-factor	0.0919	Inside Finish: Gypsum Board Cavity / Frame: R-13 / 2x4 Other Side Finish: Gypsum Board	
Roof Cons	Cathedral Ceilings	855	Wood Framed Ceiling	30	0	20	U-factor	0.0204	Roofing: Light Roof (Asphalt Shingle) Above Deck Insulation: R-20 Sheathing Roof Deck: Wood Siding/sheathing/decking Cavity / Frame: R-30 / 2x4 Sheathing / Insulation: Gypsum Board Inside Finish: Gypsum Board	N
T24-IntZB-Wall Cons	Interior Walls	78,960.1	Wood Framed Wall	0	0	0	U-factor	0.2773	Inside Finish: Gypsum Board Cavity / Frame: no insul. / 2x4 Other Side Finish: Gypsum Board	

[1] *Status: N - New, A - Altered, E - Existing*

G5. OPAQUE SURFACE ASSEMBLY SUMMARY

01	02	03	04	05	06		07	08	09	10
Surface Name	Construction Type	Area (ft^2)	Framing Type	Cavity R-Value	Continuous R-Value		Units	Value	Description of Assembly Layers	Status[1]
					Interior	Exterior				
T24-IntZB-Clg Cons	Interior Ceiling	24,291	Wood Framed Ceiling	0	0	0	U-factor	0.1957	Floor Surface: Carpeted Floor Deck: Wood Siding/sheathing/decking Cavity / Frame: no insul. / 2x12 Ceiling Below Finish: Gypsum Board	N
T24-IntZB-Flr Cons	Interior Floors	24,054	Wood Framed Floor	0	0	0	U-factor	0.1957	Floor Surface: Carpeted Floor Deck: Wood Siding/sheathing/decking Cavity / Frame: no insul. / 2x12 Ceiling Below Finish: Gypsum Board	N

[1] *Status: N - New, A - Altered, E - Existing*

G6B. OPAQUE DOOR SUMMARY (MULTIFAMILY AND COMMON AREAS)

01	02	03	04
Name	Area (ft^2)	Overall U-factor	Status[1]
Metal Door	40	0.2	N
Metal Door-2	40	0.2	N
Metal Door-2-2	40	0.2	N

[1] *Status: N - New, A - Altered, E - Existing*

G7B. FENESTRATION SUMMARY (MULTIFAMILY AND COMMON AREAS)												
01	02	03	04	05	06	07	08	09	10	11	12	13
Fenestration Name	Fenestration Type/ Product Type / Frame Type	Parent Surface	Azimuth	Multiplier	Area (ft^2)	Overall U-factor	U-factor Source	Overall SHGC	SHGC Source	Overall VT	Exterior Shading	Status[1]
Residential Window_Left _L01_1BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Left Wall_1BR	90	1	89	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L02_1BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Right Wall_1BR	270	1	139.8	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L01_2BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Left Wall_2BR	90	1	129.6	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L02_2BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Right Wall_2BR	270	1	203.7	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L01_3BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Left Wall_3BR	90	1	201.4	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L02_3BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Right Wall_3BR	270	1	316.5	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N

[1] *Status: N - New, A - Altered, E - Existing*

G7B. FENESTRATION SUMMARY (MULTIFAMILY AND COMMON AREAS)

01	02	03	04	05	06	07	08	09	10	11	12	13
Fenestration Name	**Fenestration Type/ Product Type / Frame Type**	**Parent Surface**	**Azimuth**	**Multiplier**	**Area (ft^2)**	**Overall U-factor**	**U-factor Source**	**Overall SHGC**	**SHGC Source**	**Overall VT**	**Exterior Shading**	**Status[1]**
Window (Back Combined) : CLUB ROOM/LIBRARY 01_L01	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	ExtWall (Back Combined) : CLUB ROOM/LIBRARY 01_L01	180	1	150.69	0.41	NFRC	0.26	NFRC	N/A	Standard bug screens	N
Window (Back Combined) Community Room	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	ExtWall (Back Combined) Community Room	180	1	150.69	0.41	NFRC	0.26	NFRC	N/A	Standard bug screens	N
Residential Window_Left_L01-3_1BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Left Wall-2_1BR	90	2	467.2	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left_L02-2_1BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Right Wall-2_1BR	270	2	467.2	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left_L02-2-2_1BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Back Wall-2_1BR	180	2	21.2	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left_L01-3_2BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Left Wall-2_2BR	90	2	850.8	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N

[1] *Status: N - New, A - Altered, E - Existing*

G7B. FENESTRATION SUMMARY (MULTIFAMILY AND COMMON AREAS)

01	02	03	04	05	06	07	08	09	10	11	12	13
Fenestration Name	Fenestration Type/ Product Type / Frame Type	Parent Surface	Azimuth	Multiplier	Area (ft^2)	Overall U-factor	U-factor Source	Overall SHGC	SHGC Source	Overall VT	Exterior Shading	Status[1]
Residential Window_Left _L02-2_2BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Right Wall-2_2BR	270	2	850.8	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L02-2-2_2BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Back Wall-2_2BR	180	2	38.6	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L01-3_3BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Left Wall-2_3BR	90	2	1,322	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L02-2_3BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Right Wall-2_3BR	270	2	1,322	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L02-2-2_3BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Back Wall-2_3BR	180	2	60	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L01-3-2_1BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Left Wall-2-2_1BR	90	1	116.8	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L02-2-3_1BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Right Wall-2-2_1BR	270	1	116.8	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N

[1] Status: N - New, A - Altered, E - Existing

G7B. FENESTRATION SUMMARY (MULTIFAMILY AND COMMON AREAS)

01	02	03	04	05	06	07	08	09	10	11	12	13
Fenestration Name	Fenestration Type/ Product Type / Frame Type	Parent Surface	Azimuth	Multiplier	Area (ft²)	Overall U-factor	U-factor Source	Overall SHGC	SHGC Source	Overall VT	Exterior Shading	Status[1]
Residential Window_Left _L02-2-2-2_1BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Back Wall-2-2_1BR	180	1	5.3	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L02-2-2-2-2_1BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Front Wall_1BR	0	1	5.3	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L01-3-2_2BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Left Wall-2-2_2BR	90	1	212.7	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L02-2-3_2BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Right Wall-2-2_2BR	270	1	212.7	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L02-2-2-2_2BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Back Wall-2-2_2BR	180	1	9.7	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L02-2-2-2-2_2BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Front Wall_2BR	0	1	9.7	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L01-3-2_3BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Left Wall-2-2_3BR	90	1	330.5	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N

[1] Status: N - New, A - Altered, E - Existing

G7B. FENESTRATION SUMMARY (MULTIFAMILY AND COMMON AREAS)

01	02	03	04	05	06	07	08	09	10	11	12	13
Fenestration Name	**Fenestration Type/ Product Type / Frame Type**	**Parent Surface**	**Azimuth**	**Multiplier**	**Area (ft^2)**	**Overall U-factor**	**U-factor Source**	**Overall SHGC**	**SHGC Source**	**Overall VT**	**Exterior Shading**	**Status[1]**
Residential Window_Left _L02-2-3_3BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Right Wall-2-2_3BR	270	1	330.5	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L02-2-2-2_3BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Back Wall-2-2_3BR	180	1	15	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N
Residential Window_Left _L02-2-2-2-2_3BR	Vertical fenestration Architectural Window - Operable (Multifamily only) N/A	Front Wall_3BR	0	1	15	0.3	NFRC	0.23	NFRC	N/A	Standard bug screens	N

[1] *Status: N - New, A - Altered, E - Existing*

H1. DRY SYSTEM EQUIPMENT (FURNACES, AIR HANDLING UNITS, HEAT PUMPS, VRF, ECONOMIZERS ETC.)

01	02	03	04	05	06	07	08	09	10	11	12
Equipment Name	**Equipment Type**	**Qty**	**Heating**				**Cooling**			**Economizer Type (if present)**	**Status[1]**
			Total Heating Output (kBtu/h)	**Supp Heat Output (kBtu/h)**	**Efficiency Unit**	**Efficiency**	**Total Cooling Output (kBtu/h)**	**Efficiency Unit**	**Efficiency**		
GYM_L01 AirSys	Single Zone Heat Pump (SZHP) Air System	1	36	0	COP HSPF2	3.4 6.7	36	EER2 SEER2	11.7 13.4	No Economizer	N
Leasing-Meeting Room-Office AirSys (FC1-2)	Single Zone Heat Pump (SZHP) Air System	1	36	0	COP HSPF2	3.4 7.5	36	EER2 SEER2	11.7 14.3	No Economizer	N

[1] *Status: N - New, A - Altered, E - Existing*

H1. DRY SYSTEM EQUIPMENT (FURNACES, AIR HANDLING UNITS, HEAT PUMPS, VRF, ECONOMIZERS ETC.)

01	02	03	04	05	06	07	08	09	10	11	12
			Heating				Cooling				
Equipment Name	Equipment Type	Qty	Total Heating Output (kBtu/h)	Supp Heat Output (kBtu/h)	Efficiency Unit	Efficiency	Total Cooling Output (kBtu/h)	Efficiency Unit	Efficiency	Economizer Type (if present)	Status[1]
Community Room AirSys	Single Zone Heat Pump (SZHP) Air System	1	36	0	COP HSPF2	3.4 7.5	36	EER2 SEER2	11.7 14.3	No Economizer	N

[1] *Status: N - New, A - Altered, E - Existing*

H2. DWELLING UNIT HVAC HEATING AND COOLING SYSTEMS

01	02	03	04	05	06	07	08	09	10	11	12	13
						Heating				Cooling		
Dwelling Unit Type	Equipment Name	Equipment Type	Quantity	Air Distribution System Name	Fan System name	Heat Output at 47	Heat Output at 17	Efficiency Unit	Efficiency	Total Cooling Output	Efficiency Unit	Efficiency
A1	MAESTRO OLIMPIA PRO12 HP #01925	Room HP N/A	16	N/A	N/A	12,000	N/A	HSPF2	N/A	N/A	CEER	9.9
B1	MAESTRO OLIMPIA PRO12 HP #01925	Room HP N/A	19	N/A	N/A	12,000	N/A	HSPF2	N/A	N/A	CEER	9.9
C1	MAESTRO OLIMPIA PRO12 HP #01925	Room HP N/A	19	N/A	N/A	12,000	N/A	HSPF2	N/A	N/A	CEER	9.9

H3. NONRESIDENTIAL / COMMON USE AREA FAN SYSTEMS SUMMARY

01	02	03	04	05	06	07	08	09	10	11	12	13
Name or Item Tag	Qty	Design OA CFM	Supply Fan				Return / Relief Fan					Status[1]
			CFM	Power	Power Units	Control	Fan Type	CFM	Power	Power Units	Control	
GYM_L01 AirSys	1	128.2	1,200	0.45	W/cfm	Constant Vol	N/A	N/A	N/A	N/A	N/A	N
Leasing-Meeting Room-Office AirSys (FC1-2)	1	509	1,200	0.45	W/cfm	Constant Vol	N/A	N/A	N/A	N/A	N/A	N
Community Room AirSys	1	413	1,200	0.45	W/cfm	Constant Vol	N/A	N/A	N/A	N/A	N/A	N

[1] *Status: N - New, A - Altered, E - Existing*

H3a. MULTIFAMILY / COMMON USE AREA FAN SYSTEMS SUMMARY

01	02	03	04	05
Name	Type	Power	Power Units	Status
ResidentialFanSystem 1	Fixed speed	0.45	W/cfm	N/A

H3b. MULTIFAMILY / COMMON USE AREA IAQ FAN SYSTEMS SUMMARY

01	02	03	04	05	06
Name	Type	Airflow (CFM)	Power	Power Units	Status
Corridor_L01-IAQFan	Balanced	527.2	0.7	W/cfm	N/A
Elec Room-IAQFan	Balanced	137.7	0.7	W/cfm	N/A
Corridor_L02-IAQFan	Balanced	527.2	0.7	W/cfm	N/A
Corridor_L04-IAQFan	Balanced	527.2	0.7	W/cfm	N/A
Central DU OA	Supply	2920	0.28	W/cfm	N/A

H4. MULTIFAMILY HVAC DISTRIBUTION							
01	02	03	04	05	06	07	08
Name	Type	Duct Ins. R-value Supply	Duct Ins. R-value Return	Duct Location Supply	Duct Location Return	Verified Duct Design Surface Area	
						Supply	Return
ResidentialDistributionSystem 1	No ducts (Non-Verified)	R-0.0	R-0.0			n/a	n/a
GYM_L01 AirSys- Air Dist	Conditioned space-entirely (Non-Verified)	0	0	Conditioned Zone	Conditioned Zone	0	0
Leasing-Meeting Room-Office AirSys (FC1-2)- Air Dist	Unconditioned space (Non-Verified)	0	0	Unconditioned Zone	Unconditioned Zone	0	0
Community Room AirSys- Air Dist	Unconditioned space (Non-Verified)	0	0	Unconditioned Zone	Unconditioned Zone	0	0

H9. NONRESIDENTIAL / COMMON USE AREA & HOTEL/MOTEL VENTILATION						
01	02	03	04	05	06	07
Zone Name	Mechanical Ventilation				Conditioned Area (sf)	DCV or Occupant Sensor Controls, or Both
	Ventilation Function	# of People	Supply OA CFM	Exhaust CFM		
Fitness	Sports/Entertainment - Health club/weight rooms	8.55	128.2	0	855	N/A
Leasing-Meeting Room-Office	General - Conference/meeting	33.94	509	0	1018	N/A
Community Room	General - Conference/meeting	27.53	413	0	826	N/A

H10. MULTIFAMILY DWELLING UNIT TYPE CENTRAL / INDIVIDUAL VENTILATION

01	02	03	04	05	06	07	08	09	10	11	12	13
Dwelling Unit Type	IAQ Option	Central Fan (If applicable)					Individual Fan (if applicable)					
		IAQ Fan Type Type	Supply Airflow CFM	Supply Fan Efficacy W/CFM	Exhaust CFM	Exhaust Fan Efficacy W/CFM	IAQ Fan Type	Count	Airflow CFM	Fan Efficacy W/CFM	Recovery Efficiency SRE	Recovery Efficiency ASRE
A1	Central Ventilation System - Supply	Supply	2,920	0.28	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
B1	Central Ventilation System - Supply	Supply	2,920	0.28	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
C1	Central Ventilation System - Supply	Supply	2,920	0.28	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

I2. MULTI-FAMILY WATER HEATING SYSTEM DETAIL

01	02	03	04	05	06	07	08
System Name	Configuration	Type	Qty in System	Dwelling Unit Distribution Type	Water Heater Name	Solar Heating System	Is Compact Distribution
ResidentialDHWSystem 1	Domestic Hot Water (DHW)	Central	1	Standard Distribution System	ResidentialDHWSystem 1 - heater	N/A	N/A

I3. MULTIFAMILY & HOTEL/MOTEL WATER HEATER EQUIPMENT SUMMARY - CHPWH

01	02	03	04	05	06	07	08
Name	Brand/Model	Number of Compressors	Primary Tank Volume (gal)	Tank Count	Tank R-value	Tank Location	Air Source
ResidentialDHWSystem 1	GS3-45HPA-US	9	1000	2	16	Tank Zone	Outside

I4. MULTIFAMILY & HOTEL/MOTEL WATER HEATER EQUIPMENT SUMMARY - CHPWH SECONDARY TANK								
01	02	03	04	05	06	07	08	09
Tank Type	Tank Configuration	Brand/Model	Count	Tank Volume (gal)	Tank Count	Tank R-value	Tank Location	Air Source
Electric Resistance	Series	N/A	1	200	1	16	Elec Room	N/A

I5. RECIRCULATION LOOPS					
01	02	03	04	05	06
Water Heating System Name	Number of Recirculation Loops	Loop Insulation Thickness (in)	Recirculation Loop Location	Recirculation Pump Power	Recirculation Pump Power Units
ResidentialDHWSystem 1	1	1.5	Conditioned	38	Watts

K1. INDOOR CONDITIONED LIGHTING GENERAL INFO					
01	02	03	04	05	06
Occupancy Type[1]	Conditioned Floor Area[2] (ft^2)	Installed Lighting Power (Watts)	Lighting Control Credits (Watts)	Additional (Custom) Allowance	
				Area Category Footnotes (Watts)	Area Category Footnotes (Watts)
Exercise Center Gymnasium	855	427.5	0	0	0
ConferenceMultipurposeMeetingCenterArea	1844	1383	0	0	0
Building Totals:	**2699**	**1810.5**	**0**	**0**	**0**

[1]See Table 140.6-C
[2]See NRCC-LTI--E for unconditioned spaces
[3]Lighting information for existing spaces modeled is not included in this table

K4. INDOOR CONDITIONED LIGHTING MANDATORY LIGHTING CONTROL
See NRCC-LTI-E for mandatory controls

L. DECLARATION OF REQUIRED CERTIFICATES OF INSTALLATION	
Selections made by Documentation Author indicate which Certificates of Installation must be submitted for the features to be recognized for compliance. These documents must be retained and provided to the building inspector during construction and can be found online	
Building Component	**Form/Title**
Envelope	NRCI-ENV-E - Envelope (for all buildings)
Plumbing	NRCI-PLB-E - For all buildings with Plumbing Systems
	NRCI-SAB-E - Solar Water Heating, PV and Battery Storage Systems
Indoor Lighting	NRCI-LTI-E - Indoor Lighting (for all buildings)

M. DECLARATION OF REQUIRED CERTIFICATES OF ACCEPTANCE	
Selections made by Documentation Author indicate which Certificates of Acceptance must be submitted for the features to be recognized for compliance. These documents must be provided to the building inspector during construction and must be completed through an Acceptance Test Technician Certification Provider (ATTCP).	
Building Component	**Form/Title & System Name(s)**
Envelope	NRCA-ENV-02-F - NRFC label verification for fenestration
Mechanical	NRCA-MCH-03-A - Constant Volume Single Zone HVAC
	GYM_L01 AirSys, Leasing-Meeting Room-Office AirSys (FC1-2) and Community Room AirSys., MAESTRO OLIMPIA PRO12 HP #01925
Mechanical	NRCA-MCH-20-A Multifamily Ventilation
	Central DU OA
Mechanical	NRCA-MCH-21-A Multifamily Envelope
	Dwelling Units

N. DECLARATION OF REQUIRED CERTIFICATES OF VERIFICATION
Selections made by Documentation Author indicate which Certificates of Verification must be submitted for the features to be recognized for compliance. These documents must be retained and provided to the building inspector during construction and can be found online
There are no Certificates of Verification applicable to this project

Documentation Author's Declaration Statement

1. I certify that this Certificate of Compliance documentation is accurate and complete.	
Documentation Author Name:	Documentation Author Signature:
Company:	Signature Date:
Address:	CEA/HERS Certification Identification (if applicable):
City/State/Zip: ,	Phone:

Responsible Person's Declaration statement

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Compliance is true and correct.
2. I am eligible under Division 3 of the Business and Professions Code to accept responsibility for the building design or system design identified on this Certificate of Compliance (responsible designer)
3. The energy features and performance specifications, materials, components, and manufactured devices for the building design or system design identified on this Certificate of Compliance conform to the requirements of Title 24, Part 1 and Part 6 of the California Code of Regulations.
4. The building design features or system design features identified on this Certificate of Compliance are consistent with the information provided on other applicable compliance documents, worksheets, calculations, plans and specifications submitted to the enforcement agency for approval with this building permit application.
5. I understand that a registered copy of this Certificate of Compliance shall be made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to accomplish this requirement.
6. I understand that a registered copy of this Certificate of Compliance is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to accomplish these requirements.

Responsible Designer Name:	Responsible Designer Signature:	
Company:		
Address:	Date Signed:	
City/State/Zip: ,	License #:	
Phone:	Title:	Scope: